

Lab 9

Description. This lab aims to introduce you to bash shell scripting. During this lab, you will learn the principle of bash scripting. You will write and execute your first bash script on a GNU/Linux operating system.

Task 1. In this first task, the instructor will use some slides to introduce operating systems shells and shell scripting. They will also present the language used for writing GNU bash scripts so that you can start writing your own scripts.

Task 2. In this second task, you will start writing and executing your first scripts:

2.1. Write a bash script that creates a folder named `OS_Lab9` and then creates 150 txt files, named `file_x`, where $x \in \{1 \dots 150\}$.

2.2. Write a bash script that computes the sum of the parameters provided as arguments to the script. The number of arguments is not limited.

2.3. Write a bash script `text_stats.sh` to perform some basic statistics on text files located in a specific directory. First, create a new directory `text_files` in your home directory. Then, download some text files from Project Gutenberg and save them in the `text_files` directory. You can use the following commands to download some files:

```
$cd /text_files
$wget https://www.gutenberg.org/files/11/11-0.txt
$wget https://www.gutenberg.org/files/84/84-0.txt
```

Your script takes the following arguments: the directory containing the text files to be analyzed and the word to be searched for in the text files. When the script is run with the correct arguments, it should perform the following statistics on the text files in the directory: Provide the total number of words in all the files, and provides the total number of times the specified word appears in all the files.

☞ Use the following command to search and count the number of times the specified word appears in a specific file:

```
grep -o -w "$word" "$file" | wc -l
```

☞ Use the following command to count the total number of words in all files:

```
n_w=$(find "$dir" -type f | xargs wc -w | tail -n 1 | awk '{print $1}')
```

2.4. Write a bash script that asks the user to guess an integer value between 0 and 100. The script gives the user 10 tries to guess the right value.

Note. If you do not have a Unix-like OS, you may want to use an online bash interpreter at https://www.onlinegdb.com/online_bash_shell.